Tail-Waste Expressions

$$F_{\text{exact}}(C) = 100 \cdot \frac{\left\lceil \frac{C}{T} \right\rceil T - C}{\left\lceil \frac{C}{T} \right\rceil T}$$

$$F_{\text{worst}}(C) \approx 100 \cdot \frac{T}{C + T}$$

Variable meanings

- C: context length in tokens for the request.
- T: number of cached tokens that fit in one physical page.
- $\left\lceil \frac{C}{T} \right\rceil$: number of pages mapped for context length C.
- $\left\lceil \frac{C}{T} \right\rceil T$: number of tokens that could be held by all mapped blocks
- $\left\lceil \frac{C}{T} \right\rceil T - C$: unused token capacity in the final mapped page, i.e., tail waste.
- $F_{\text{exact}}(C)$: exact tail-waste percentage at context length C.
- $F_{\text{worst}}(C)$: smooth worst-case upper envelope of the fragmentation peaks.

Tokens Per Page by Architecture

$$T_{\text{dense}} = \left\lfloor \frac{P}{H_{\text{kv,loc}} D b} \right\rfloor$$

$$T_{\text{MLA}} = \left\lfloor \frac{P}{(r_{\text{kv}} + d_{\text{rope}}) b} \right\rfloor$$

Sources of the terms

- P: physical page size in bytes. In our experiments, this is fixed at 2 MB.
- $H_{\text{kv,loc}}$: local KV-head count on one tensor-parallel worker, taken from the model architecture and parallel configuration.
- D: per-head hidden dimension, taken from the trained model configuration.
- b: bytes per stored cache element, determined by the KV cache datatype such as FP16 or BF16.
- r_{kv}: KV latent rank used by MLA, taken from the MLA model configuration.
- d_{rope}: dimensionality of the RoPE-retained key component in MLA, also taken from the MLA model configuration.
- $\lfloor \cdot \rfloor$: indicates that only a whole number of token slots can fit in one page.
- T_{dense}: dense-layout tokens per page for MHA or GQA.
- T_{MLA}: MLA-layout tokens per page.